

Lab 7: Trees & Recursion

CSCI 41

Objectives:

- Gain a better understanding of recursion
- Implement a ton of recursive functions on trees

Part 1: Get the starter code

First, `cd` into the folder where you want to store your class files (e.g., `~/labs`). Copy the starter code there with the following line:

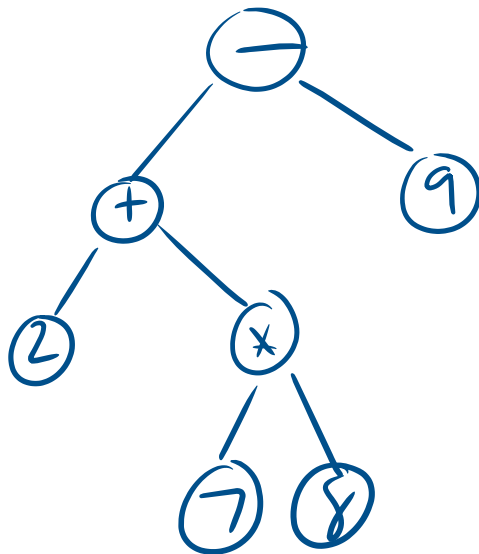
```
gimme lab07@csci41
```

This will make a `lab07` folder in whatever directory you were in.

Part 2: Implement a bunch of recursive tree functions

Part 2.0: Understand expression trees

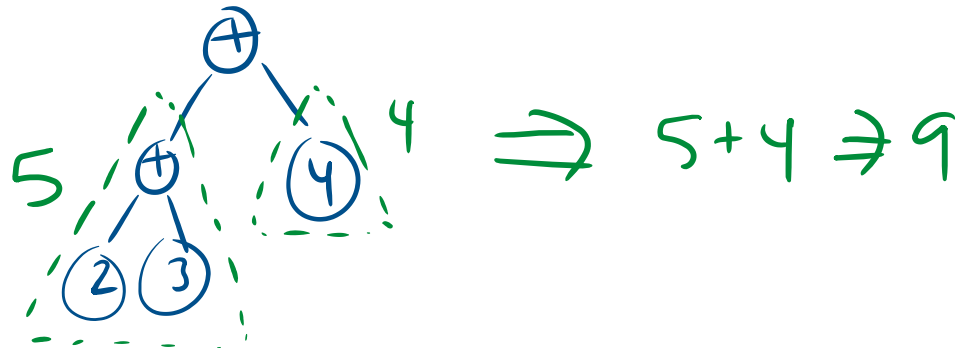
An expression tree for the expression $(2 + (7 * 8)) - 9$ looks like this:



Operators are always at the interior nodes, and numbers are always at the leaves.

Part 2.1: Evaluate expression trees recursively

Given an expression tree, you can recursively *evaluate* it to compute an answer. See the following diagram:

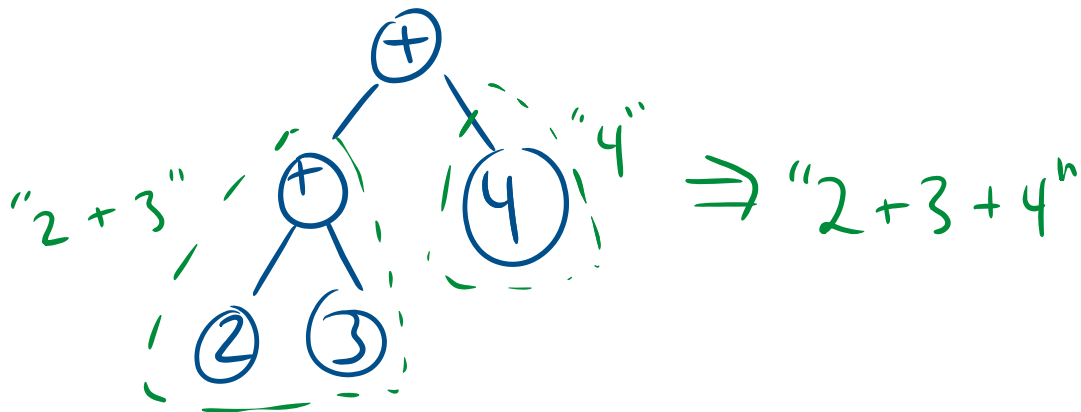


The recursive case should evaluate the expression in each subtree.

Base case hint: every leaf is just a single number—that number is their result!

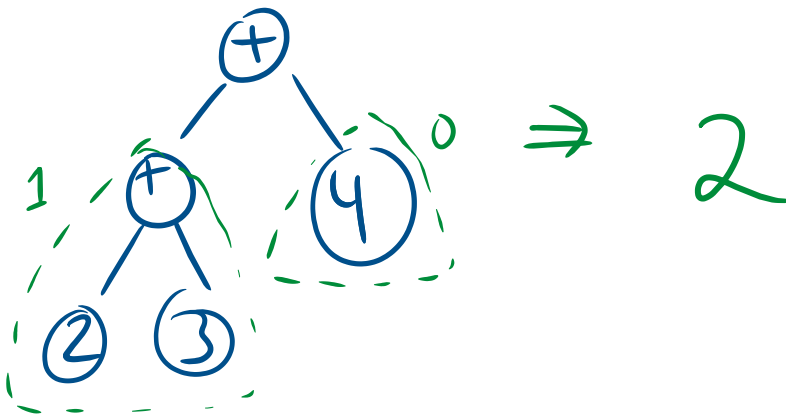
Part 2.2: {pre, post, in}order, breadth first

These are the same as what I presented in class, except this time you're returning a string instead of printing anything out. See this diagram for an example of what inorder should return:



Part 2.3: Height

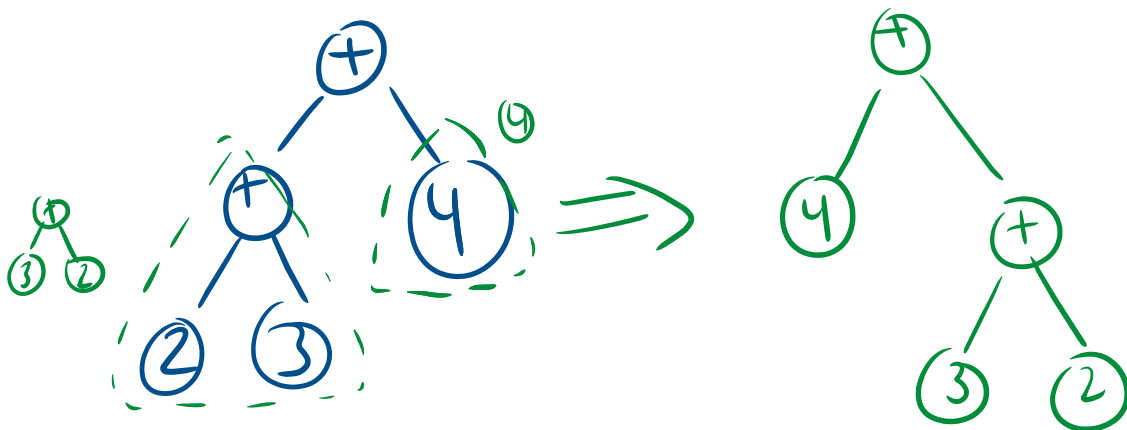
The height of a tree is the longest number of edges from the root to a leaf node. Here's an example:



The height of an empty tree is defined to be -1.

Part 2.4: Reverse (Mirror)

To reverse a tree, you flip all the left and right subtrees. Here's a picture of what your code should do:



This is a void function, so remember to *modify the tree that was given to you*.

Compile the tests with `make expressionsTests` and get them to pass. You don't have to make any of your own tests this time.

Part 3: Submit your code

When you're satisfied with your solutions, you can submit them to the autograder for grading. **Do not submit and assume you got 100%—you may be unpleasantly surprised. Always check your grade.**

From your solution directory, submit your code to the autograder using the following command on the terminal:

```
turnin lab07@csci41 expressions.cpp
```

The autograder will grade your code within a minute or so. If it's not working, please yell at Lawton to fix it. Run `view-grades` to look at your grades on the terminal, or go to the class website to view them there. You may resubmit as much as you want before the due date—just follow this same process after you've updated your programs.

Rubric

Rubric Item	Points (32 pts total)
Lawton's tests pass	32 pts (2 for each test)